

Web Development: An Overview

Web development is the work involved in building and maintaining websites and web applications that run in a browser.

It is broadly split into front-end development, which shapes what users see, and back-end development, which handles logic and data.

Front-end development relies on three core technologies: HTML for structure, CSS for styling, and JavaScript for interactivity.

HTML is used to structure the content of a page, defining elements such as headings, paragraphs, links, and images.

CSS controls the visual presentation of a page, including layout, colors, fonts, and responsive behavior across screen sizes.

JavaScript allows a web page to respond to user actions, update content dynamically, and communicate with a server without reloading.

Modern front-end development often uses frameworks such as React, Vue, or Angular to build complex, interactive interfaces.

Back-end development handles the server side of a web application, including business logic, databases, and authentication.

Many back-end services are written in languages such as JavaScript, Java, Ruby, or Python, using frameworks like Express or Django.

Python-based frameworks such as Django and Flask are popular choices for building back-end APIs and full web applications.

APIs, or application programming interfaces, define how the front end and back end communicate, often using JSON over HTTP.

A web server receives incoming requests, processes them, and sends back a response, often rendered HTML or structured data.

Databases store the persistent data behind a web application, from user accounts to product catalogs and order history.

Version control systems such as Git are essential for tracking changes to a project's code over time.

Deployment is the process of making a web application available on the internet, often using cloud platforms or hosting providers.

Responsive design ensures that a website looks and functions well on devices ranging from large monitors to small phones.

Web accessibility focuses on making sites usable by people with disabilities, including those using screen readers or keyboard navigation.

Performance optimization, such as reducing file sizes and minimizing server requests, keeps pages loading quickly for users.

Security practices, including input validation and protection against common attacks, are a critical part of any web project.

As the web continues to evolve, developers must keep learning new tools while relying on the same core fundamentals.